

Question 3 Convolutional Architectures (16 points)

Consider the convolutional neural network defined by the layers in the left column below. Fill in the shape of the output volume and the number of parameters at each layer. You can write the activation shapes in the format (H, W, C) , where H, W, C are the *height*, *width* and *channel* dimensions, respectively. Unless specified, *assume padding 1, stride 1 where appropriate*.

Notation:

- CONV x - N denotes a convolutional layer with N filters with height and width equal to x .
- POOL- n denotes a $n \times n$ max-pooling layer with stride of n and 0 padding.
- FLATTEN flattens its inputs, identical to `torch.nn.flatten` / `tf.layers.flatten`
- FC- N denotes a fully-connected layer with N neurons

Layer	Activation Volume Dimensions	Number of parameters
Input	$32 \times 32 \times 3$	0
CONV3-8		
Leaky ReLU		
POOL-2		
BATCHNORM		
CONV3-16		
Leaky ReLU		
POOL-2		
FLATTEN		
FC-10		

Solution:

Layer	Activation Volume Dimensions	Number of parameters
Input	$32 \times 32 \times 3$	0
CONV3-8	$32 \times 32 \times 8$	$8 \cdot (3 \cdot 3 \cdot 3 + 1) = 224$
Leaky ReLU	$32 \times 32 \times 8$	0
POOL-2	$16 \times 16 \times 8$	0
BATCHNORM	$16 \times 16 \times 8$	$2 \cdot 8$
CONV3-16	$16 \times 16 \times 16$	$16 \cdot (3 \cdot 3 \cdot 8 + 1) = 1168$
Leaky ReLU	$16 \times 16 \times 16$	0
POOL-2	$8 \times 8 \times 16$	0
FLATTEN	$16 \cdot 8 \cdot 8$	0
FC-10	10	$(8 \cdot 8 \cdot 16 + 1) \cdot 10 = 10250$